



SYDE252 - W02A - Complex Signals
and the Unit Impulse
Dr. Matt Borland

<https://en.wikipedia.org/wiki/Buddhabrot>

What are we doing today:

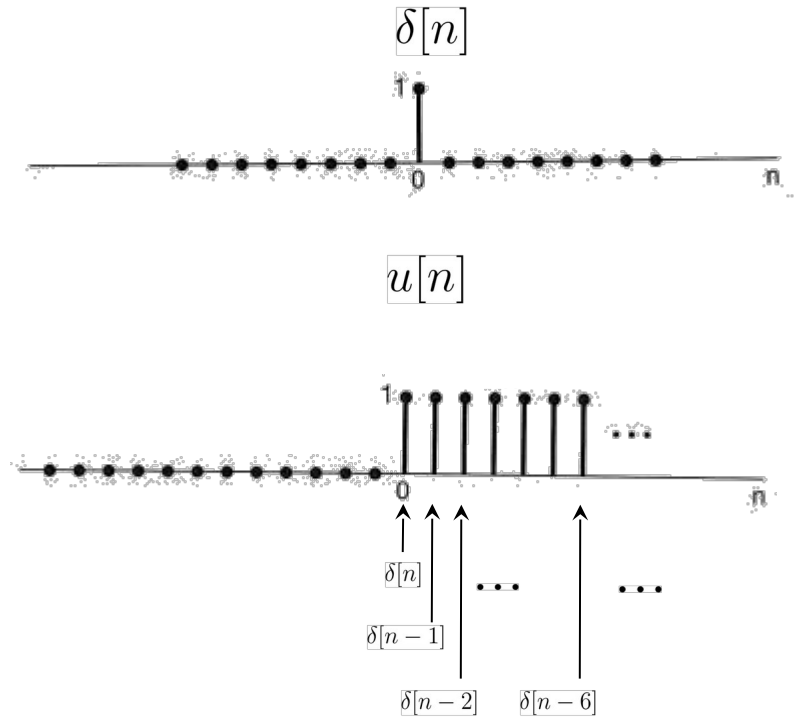
- Introducing two representations of ideal signals
 - Complex exponentials
 - Unit Impulse
- Both have definitions in CT and DT

1.4 Unit impulse and unit step

DT unit impulse and unit step

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



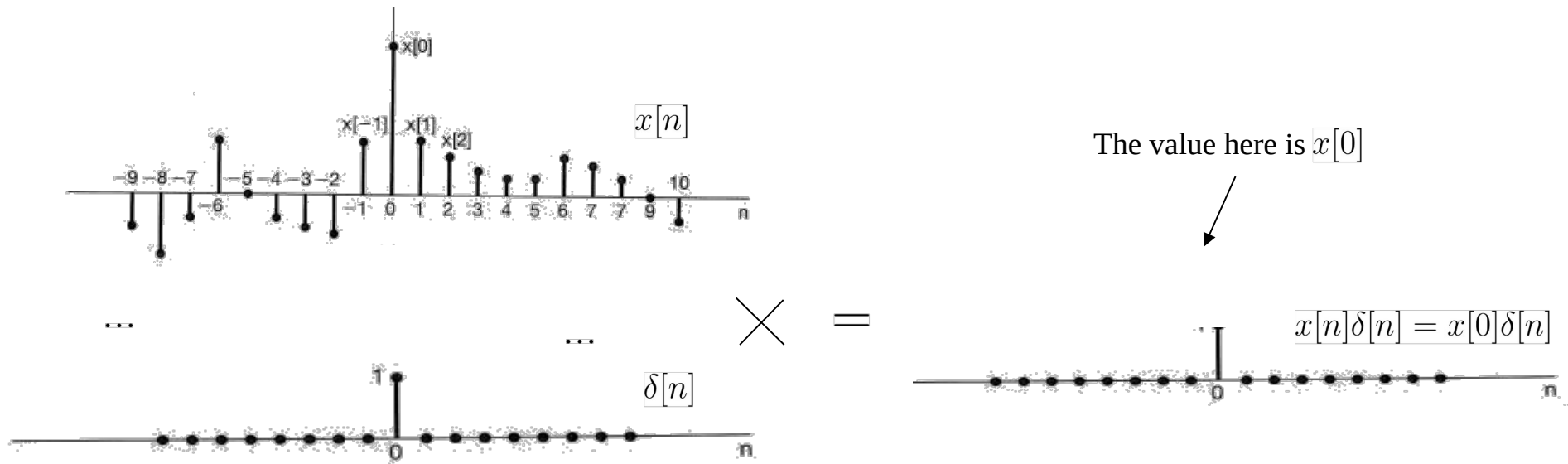
$$u[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \dots + \delta[n-k] + \dots,$$

$$= \sum_{k=0}^{+\infty} \delta[n-k]$$

Unit impulse and unit step

DT unit impulse and unit step

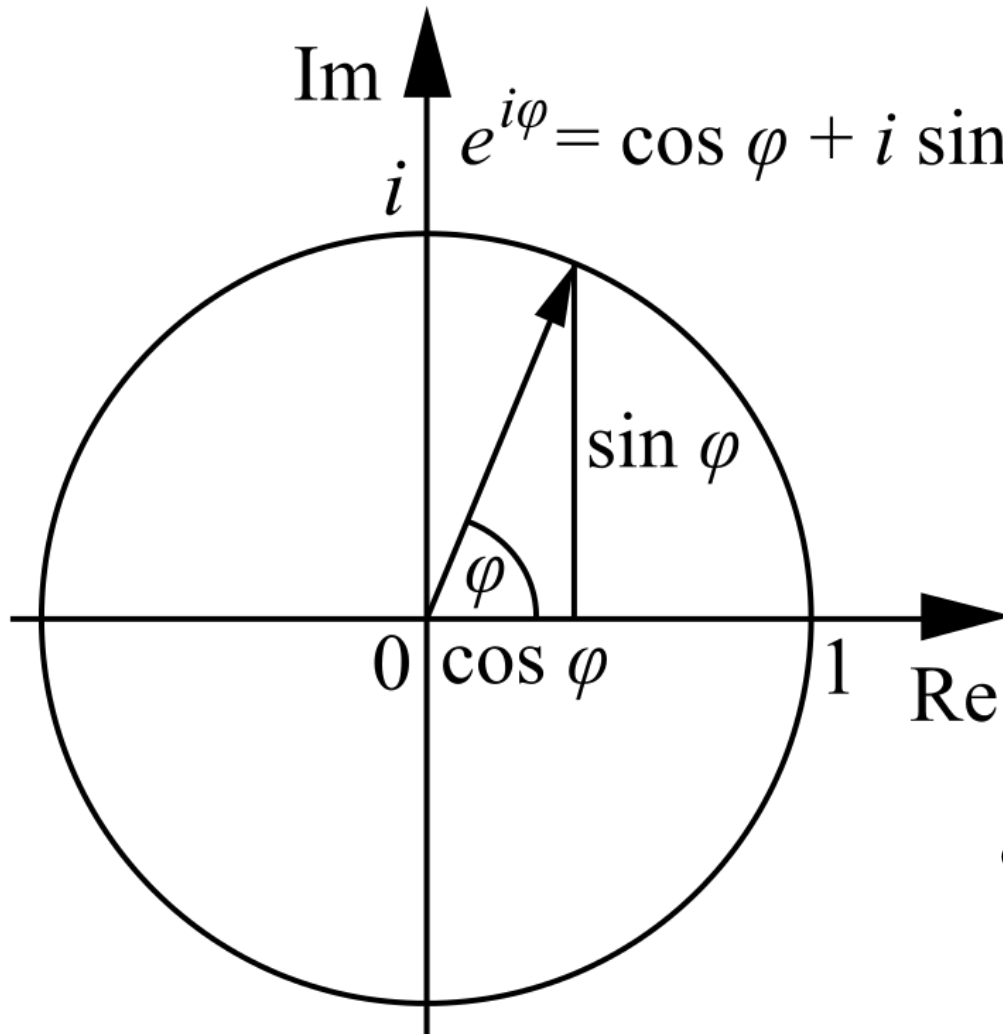
Sampling property of the DT unit impulse function:



$$x[n]\delta[n] = \dots + x[-2]\delta[-2] + x[-1]\delta[-1] + x[0]\delta[0] + x[1]\delta[1] + x[2]\delta[2] + \dots$$

$\dots$ 0 0 1 0 0 $\dots$

$\delta[n]$ is also called **unit sample** (function)



Euler's Formula!

$$\cos(x) = \operatorname{Re}(e^{ix}) = \frac{e^{ix} + e^{-ix}}{2}$$

$$\sin(x) = \operatorname{Im}(e^{ix}) = \frac{e^{ix} - e^{-ix}}{2i}$$

https://en.wikipedia.org/wiki/Euler%27s_formula

Complex exponential signals

CT Complex exponential signals

$$\mathcal{E}\{e^{j\omega_0 t}\} = \cos(\omega_0 t) \quad \mathcal{O}\{e^{j\omega_0 t}\} = j \sin(\omega_0 t)$$

$$e^{j\omega_0 t} = \mathcal{E}\{e^{j\omega_0 t}\} + \mathcal{O}\{e^{j\omega_0 t}\} = \cos(\omega_0 t) + j \sin(\omega_0 t)$$

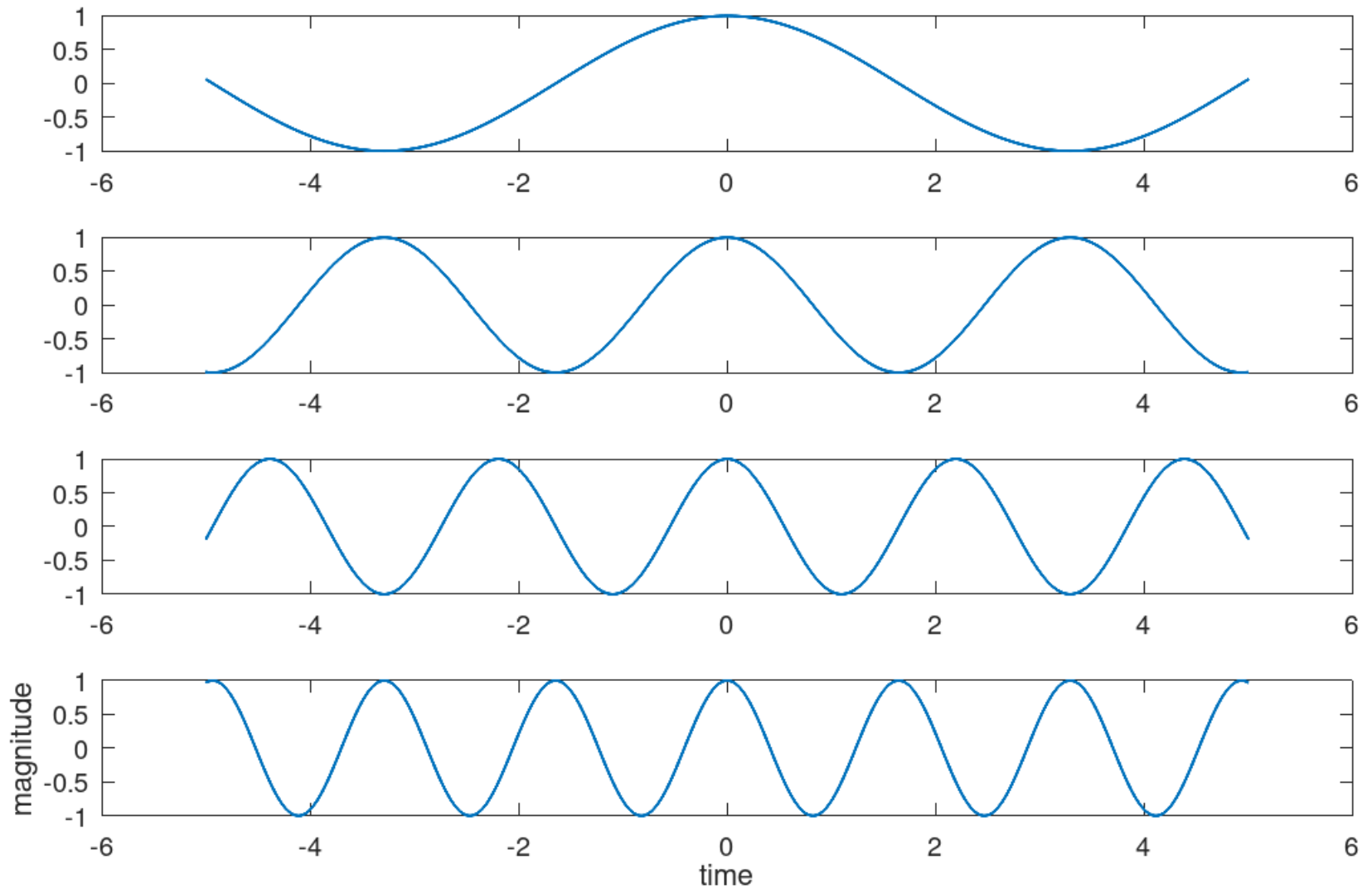
Both parts are periodic, therefore $e^{j\omega_0 t}$ is periodic.

The fundamental period is $T_0 = \frac{2\pi}{\omega_0}$ (s), and the fundamental frequency is ω_0 (radians / s)

It follows that $e^{j2\omega_0 t}, e^{j3\omega_0 t}, \dots, e^{jk\omega_0 t}, \dots$ are all periodic.

	Fund. freq. (rad/s)	Fund. period (s)	
$e^{j\omega_0 t}$	ω_0	$2\pi/\omega_0$	Harmonically related complex exponentials: A set of complex exponentials whose fund. freq. are all multiples of a single frequency.
$e^{j2\omega_0 t}$	$2\omega_0$	π/ω_0	
$e^{j3\omega_0 t}$	$3\omega_0$	$2\pi/3\omega_0$	
$e^{jk\omega_0 t}$	$k\omega_0$	$2\pi/k\omega_0$	

Harmonics



CT Complex exponential signals

Several notes on CT complex exponential signals $e^{j\omega_0 t}$

- Have a central role in our analysis of CT signals and systems
- **Always periodic** regardless the value of ω_0
- The **larger** the value of ω_0 , the **higher the frequency** of the signal.

DT Complex exponential signals

$$x[n] = e^{j\omega_0 n} = \cos(\omega_0 n) + j \sin(\omega_0 n)$$

Is $x[n]$ periodic, like the CT complex exponential signals?

We need to find an N , such that $x[n + N] = x[n]$

Plug in the signal, we have $e^{j\omega_0(n+N)} = e^{j\omega_0 n} e^{j\omega_0 N}$

In order to get the above condition, we need $e^{j\omega_0 N} = 1$

Therefore, we need $\omega_0 N = 2\pi m$, **where N and m are integers.**

Or we need $\frac{2\pi}{\omega_0} = \frac{N}{m}$. In other words, $\frac{2\pi}{\omega_0} \in \mathcal{Q}$ (**rational**)

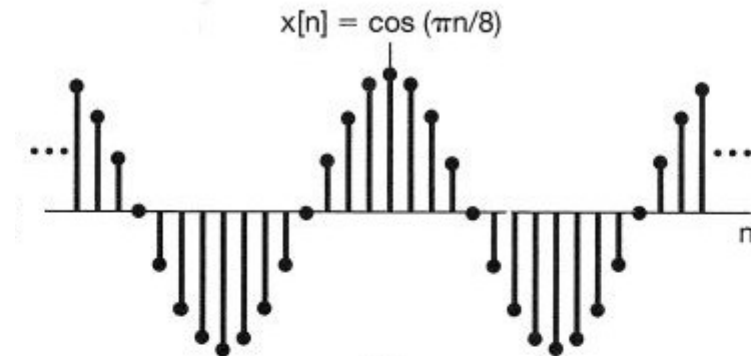
DT Complex exponential signals

Example 1 $x[n] = e^{j\frac{\pi}{8}n}$

$$\omega_0 = \frac{\pi}{8} \Rightarrow \frac{2\pi}{\omega_0} = 16 \Rightarrow N = 16, m = 1$$

Therefore, the signal is **periodic**;

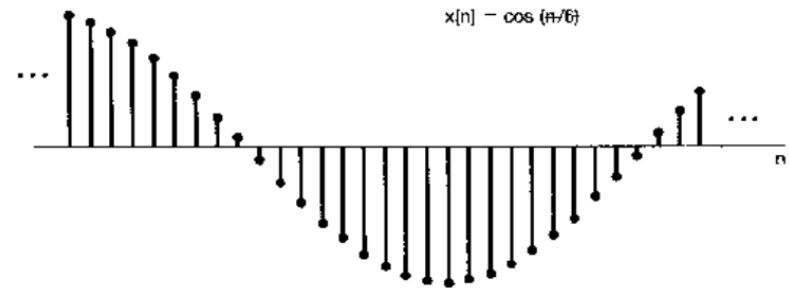
It's fund. period is $N=16$, fund. freq. is $\frac{\pi}{4}$ rad/s



Example 2 $x[n] = e^{j\frac{1}{6}n}$

$$\omega_0 = \frac{1}{6} \Rightarrow \frac{2\pi}{\omega_0} = 12\pi \quad \text{not rational}$$

Therefore, the signal is **NOT periodic**;



$$\omega_0 N = 2\pi m \quad \text{where } N \text{ and } m \text{ are integers.}$$

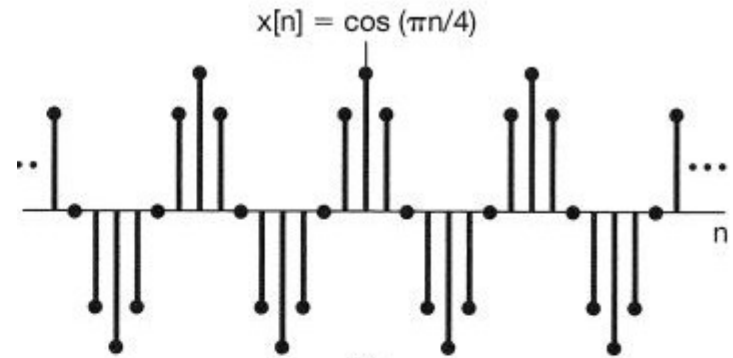
DT Complex exponential signals

Example 3 $x[n] = e^{j\frac{\pi}{4}n}$

$$\omega_0 = \frac{\pi}{4} \Rightarrow \frac{2\pi}{\omega_0} = 8 \Rightarrow N = 8, m = 1$$

Therefore, the signal is **periodic**;

It's fund. period is $N=8$, fund. freq. is $\frac{\pi}{4}$ rad/s



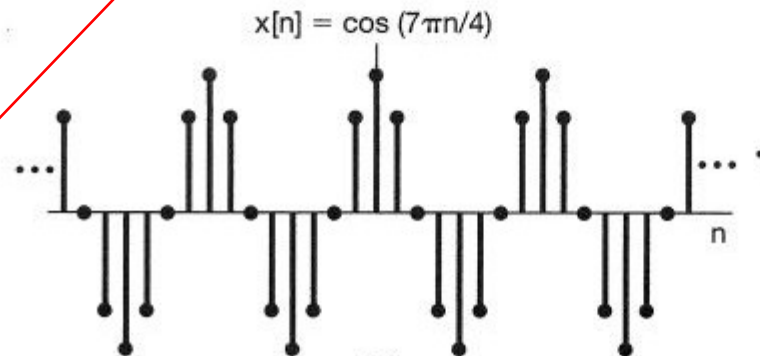
The two signals, with different ω_0 , have the same freq.

Example 4 $x[n] = e^{j\frac{7\pi}{4}n}$

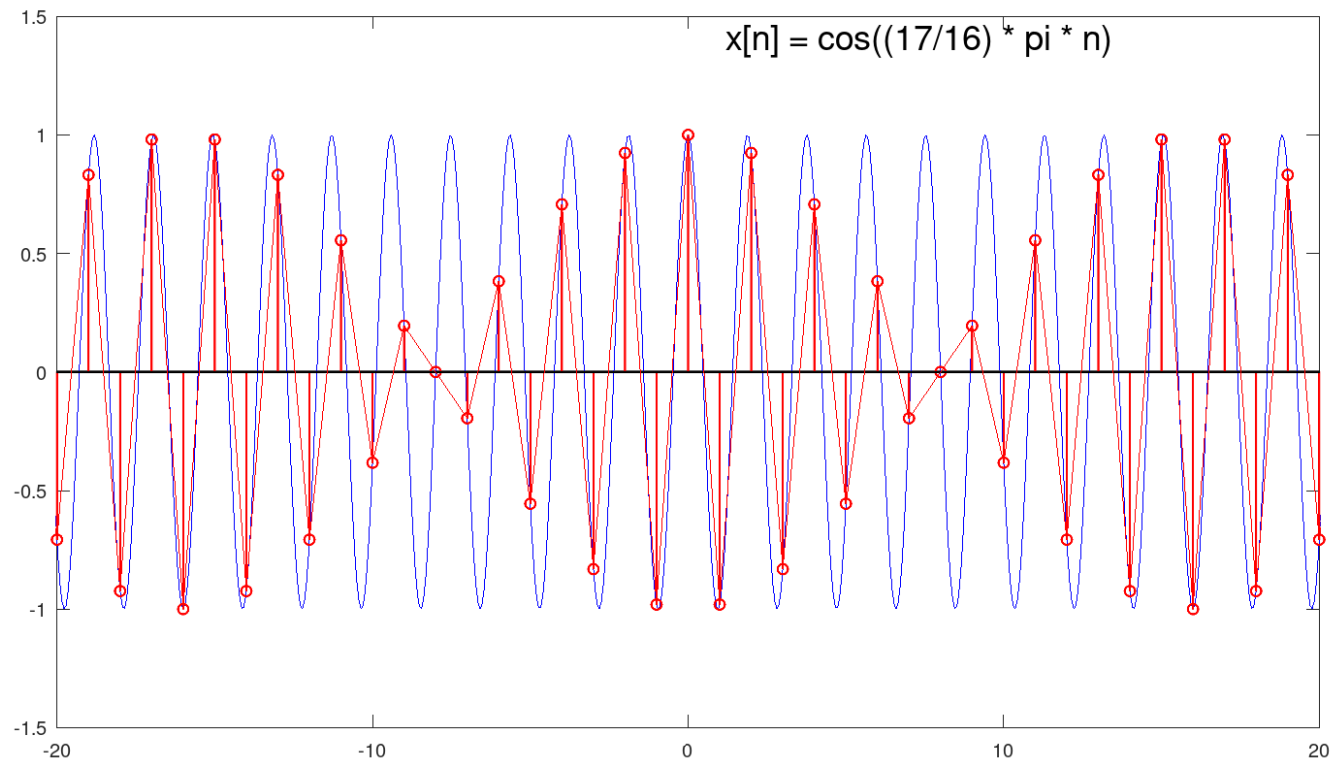
$$\omega_0 = \frac{7\pi}{4} \Rightarrow \frac{2\pi}{\omega_0} = \frac{8}{7} \Rightarrow N = 8, m = 7$$

Therefore, the signal is **periodic**;

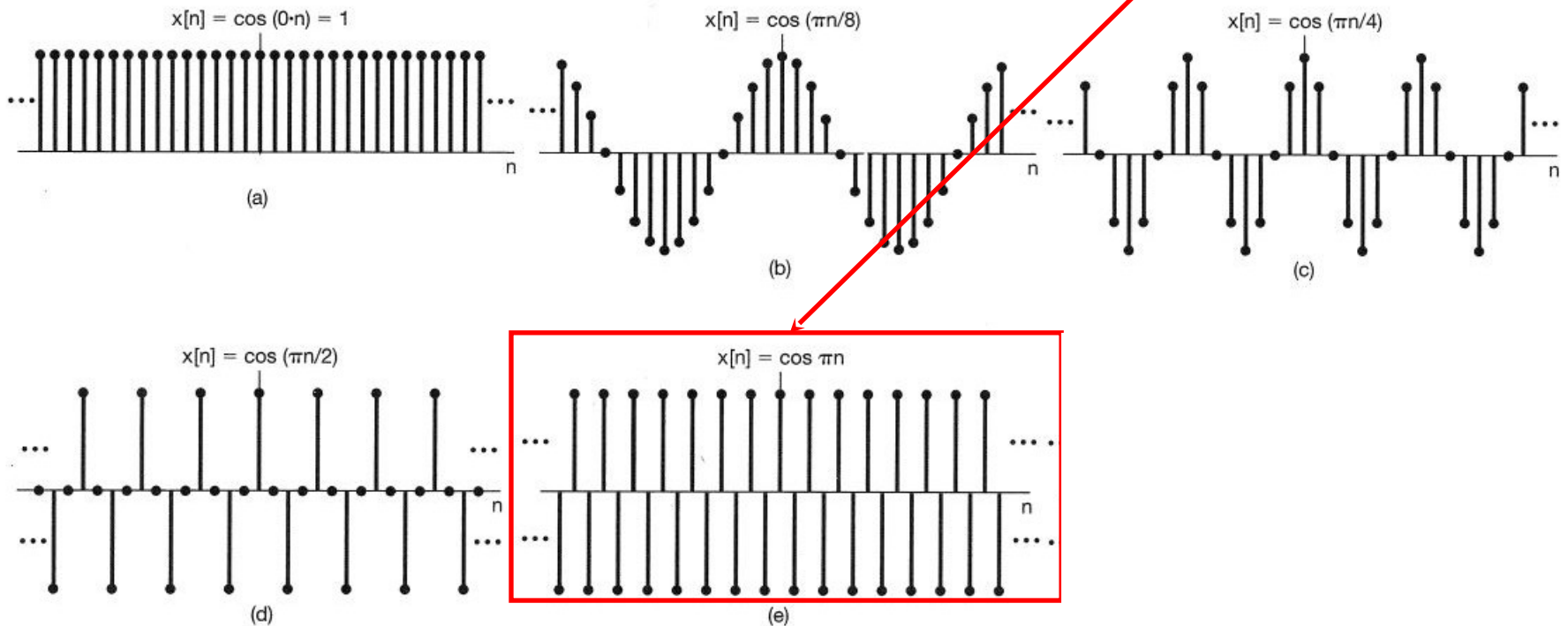
It's fund. period is $N=8$, fund. freq. is $\frac{\pi}{4}$ rad/s



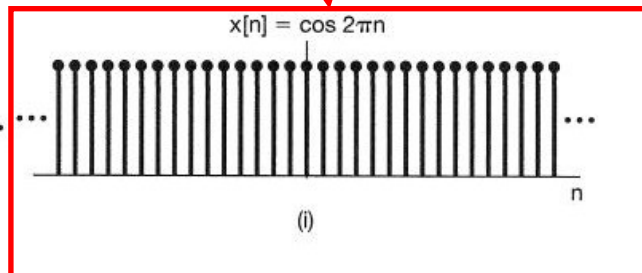
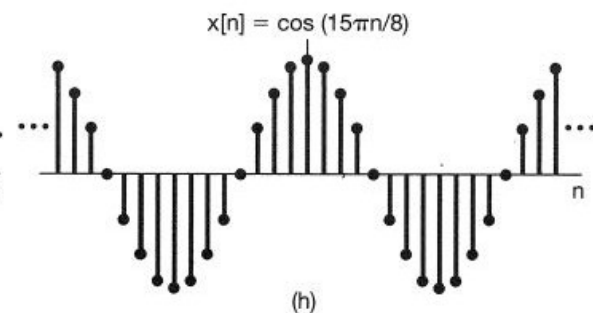
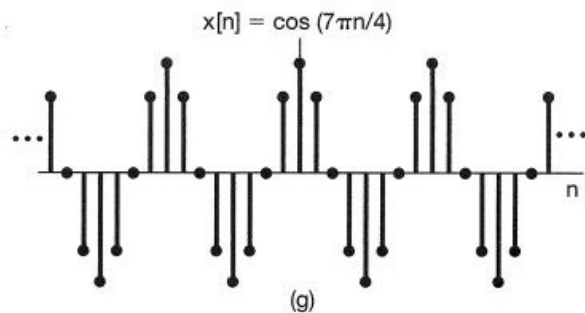
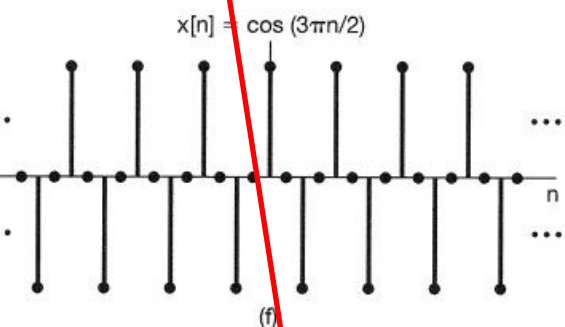
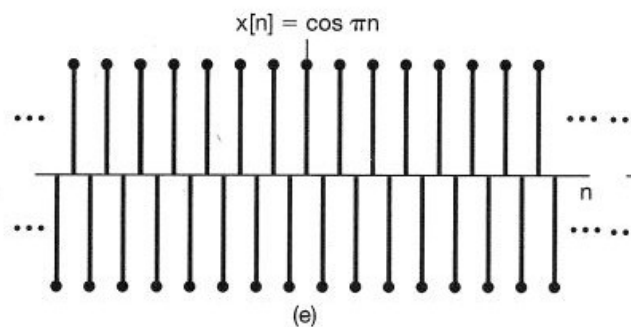
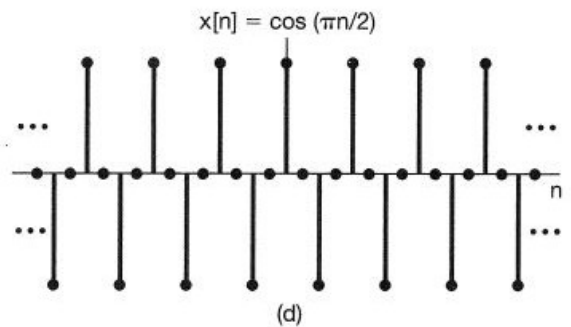
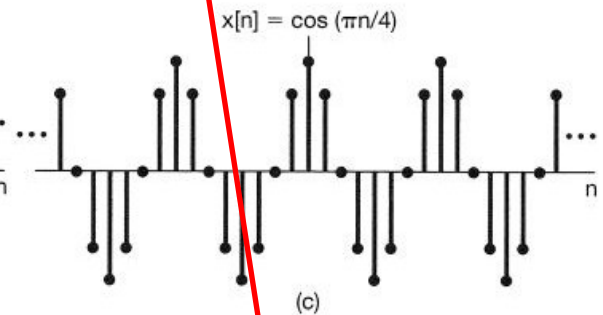
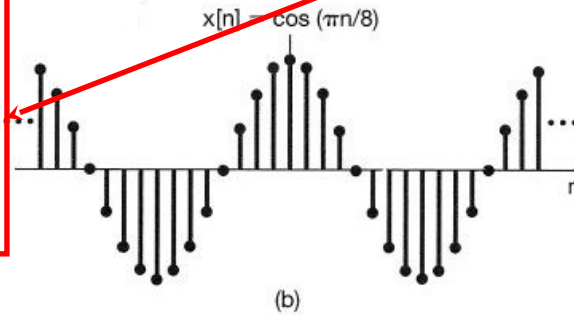
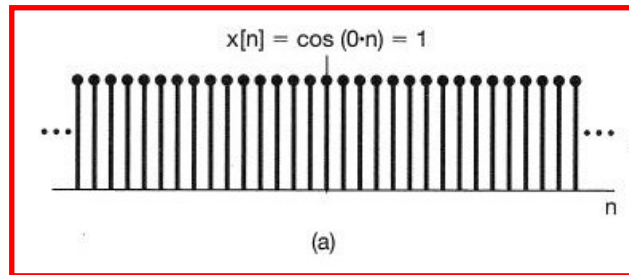
DT Sinusoids don't always “fit” on the grid...



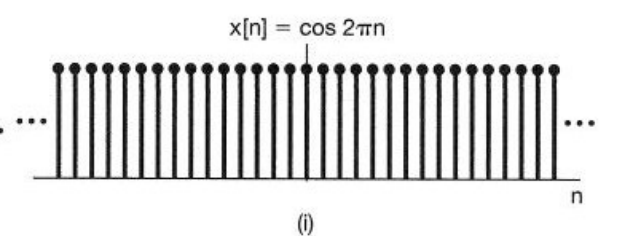
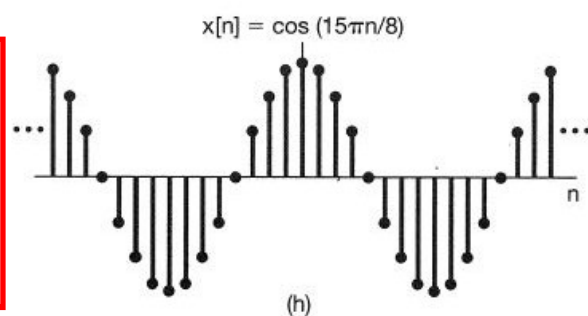
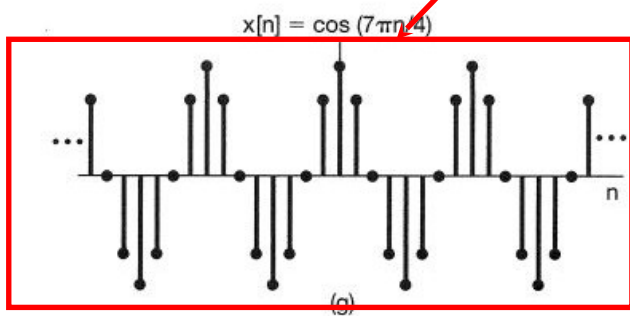
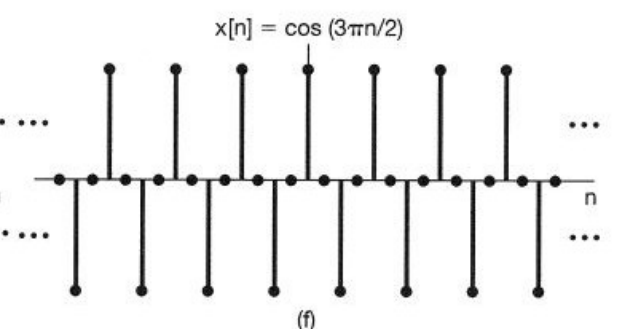
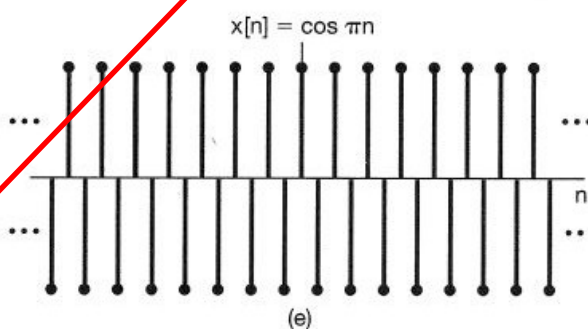
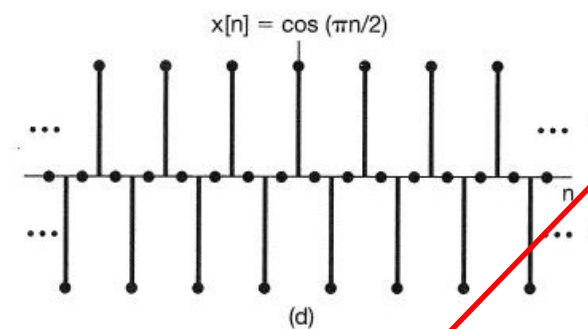
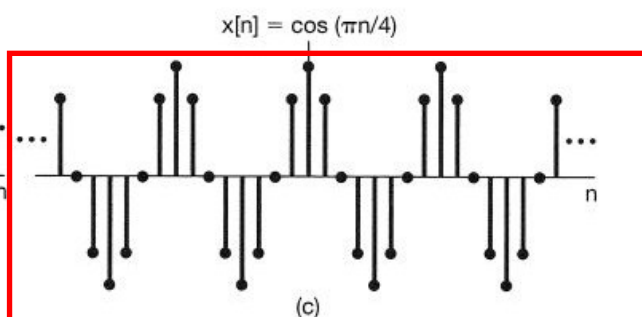
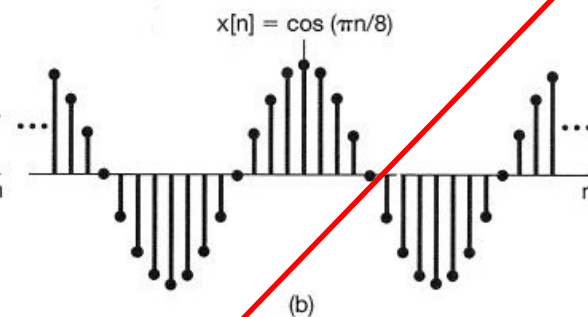
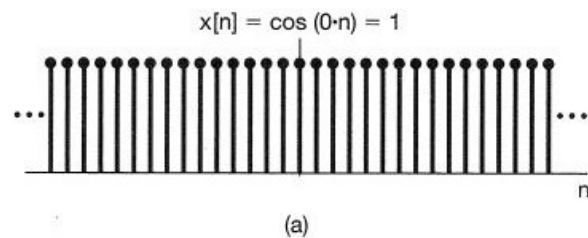
$\omega_0 = \pi$, the signal has **the fastest change**:
the signal takes max. and min. values at two adjacent values of n



$\omega_0 = 0, 2\pi$, the signal has the slowest change:
a constant value



Different ω_0 , but the signals are identical.



Several notes on DT complex exponential signals $e^{j\omega_0 n}$

- Have a central role in our analysis of DT signals and systems
- It is periodic **ONLY** when $\frac{2\pi}{\omega_0} = \frac{N}{m} \in \mathcal{Q}$ (**a rational number**)

In this case, fund. period is N , and fund. freq. is $\frac{\omega_0}{m}$.

- The fund. frequency of the signal does **NOT** monotonically increase with ω_0 .
- When ω_0 is the **odd multiple of π** , the signal has the **highest frequency**;
- When ω_0 is the **even multiple of π** , the signal has the **lowest frequency**;